

Original Communication

Effect of Coenzyme Q10 on Oxidative Stress, Glycemic Control and Inflammation in Diabetic Neuropathy: A Double Blind Randomized Clinical Trial

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Received: May 29, 2014; Accepted: January 12, 2015

Abstract: Objective: This 12-week randomized placebo controlled clinical trial investigated the effect of Coenzyme Q10 (CoQ10) on diabetic neuropathy, oxidative stress, blood glucose and lipid profile of patients with type 2 diabetes. Methods: Diabetic patients with neuropathic signs (n = 70) were randomly assigned to CoQ10 (200 mg/d) or placebo for 12 weeks. Blood samples were collected for biochemical analysis and neuropathy tests before and after the trial. Results: There were no significant differences between the two groups in terms of mean fasting blood glucose, HbA1c and the lipid profile after the trial. The mean insulin sensitivity and total antioxidant capacity (TAC) concentration significantly increased in the Q10 group compared to the placebo after the trial ($P < 0.05$). C-reactive protein (hsCRP) significantly decreased in the intervention group compared to placebo after the trial ($P < 0.05$). In the control group, insulin sensitivity decreased and HOMA-IR increased, which revealed a significant difference between groups after the trial. Neuropathic symptoms and electromyography measurements did not differ between two groups after the trial. Conclusions: According to the present study, CoQ10, when given at a dose of 200 mg/d for 12 weeks to a group of neuropathic diabetic patients, did not improve the neuropathy signs compared to placebo, although it had some beneficial effects on TAC and hsCRP and probably a protective effect on insulin resistance.

Key words: diabetic neuropathy, oxidative stress, blood glucose, lipid profile, insulin sensitivity

Introduction

Type 2 diabetes is a clinical syndrome with variable phenotypic expression rather than a single disease with a specific etiology. The main etiology of the syndrome includes β -cell insufficiency and insulin resistance, which leads to increased blood glucose. High blood glucose level determines the overproduction of reactive oxygen species (ROS) by the mitochondria electron transport chain. High reactivity of ROS determines chemical changes in virtually all cellular components, leading to DNA and protein modification and lipid peroxidation [1]. One of the chief injuries arising from hyperglycemia is injury to vasculature, which is classified as either small vascular injury (microvascular disease) including retinopathy, nephropathy and neuropathy, or injury to the large blood vessels of the body (macrovascular disease) [2]. Diabetic peripheral neuropathy (DPN) is one of the most prevalent long-term complications of diabetes. More than 50 % of all diabetic patients may suffer from some degree of neuropathy [3]. DPN is considered the cause of considerable morbidities and can affect the quality of life [3, 4]. It is characterized by the progressive deterioration of nerves predisposing neuropathic foot ulceration, Charcot neuroarthropathy, and lower extremity amputation [4]. Diabetic neuropathies are divided into symmetrical and asymmetrical types; symmetrical forms include distal sensory or sensory polyneuropathy, small-fiber neuropathy, autonomic neuropathy and large-fiber neuropathy [5]. Older age, long duration of diabetes and poor glycemic control are well established risk factors for DPN [6]. Chronic hyperglycemia causes oxidative stress in tissues susceptible to complications in diabetic patients. The mechanisms underlying oxidative stress in chronic hyperglycemia and neuropathy development have been studied in experimental models [7]. As a result, ameliorating oxidative stress through treatment with antioxidants might be an effective strategy for the reduction of DPN [8].

Coenzyme Q10 is a quinone which was first isolated from bovine heart mitochondria. It is also known as ubiquinone, because it is found in virtually all human cells. The reduced form of Coenzyme Q10 acts as an antioxidant, combats free radicals, prevents lipid peroxidation, and protects mitochondrial DNA. Coenzyme Q10 has been suggested to increase plasma antioxidant activity [9].

The effect of Coenzyme Q10 on oxidative diseases such as diabetes, coronary artery disease and hypertension has been studied [10–12]. There are limited data regarding the effect of Coenzyme Q10 on diabetic

neuropathy [13] and oxidative stress. Therefore, the aim of this study was to investigate the effect of Coenzyme Q10 supplementation on oxidative stress in a group of diabetic patients suffering from neuropathy.

Materials and Methods

The subjects for this randomized, double-blind, placebo-controlled, parallel group study were recruited from Yazd Diabetes Research Center, Iran. The trial has been done from October 2011 to February 2012 (RCT code: IRCT201109127541N1) and was planned for 12 weeks (Figure 1).

The study protocol was approved by the Ethics Committee of Shahid Sadoughi University of Medical Sciences, Yazd, Iran. The sampling was performed by randomizing patients who fulfilled our inclusion criteria. All participants were referred to a single endocrinologist. Subjects who were recruited for the trial (blinded to group assignment) were informed about the aims, procedures and possible risks of the study and gave written informed consent. The inclusion criteria were age between 35 and 65 years, type 2 diabetes defined by the American Diabetes Association criteria (1997), diabetes duration > 5 years, Michigan Neuropathy Screening Instrument (MNSI) score ≥ 8 , impaired knee and Achilles reflex, abnormal nerve conduction velocity and on a stable dose of medications for diabetic control in the month prior to enrolment. The patients should not have taken antioxidant supplements during the last three months. Subjects with liver, kidney or other neurologic diseases were excluded.

Participants were randomly allocated in a 1:1 ratio to receive the supplement or matched placebo daily for 12 weeks. After randomization, patients received an unmarked bottle of capsules with either 100 mg CoQ10 (Health Burst, USA) or the placebo. They were instructed to take CoQ10 or placebo capsules twice daily with their meals, and to leave unused capsules in the bottles. Participants were instructed to follow their habitual diet and physical activity and not to change their prescribed medications and dosage. The placebo capsule contents consisted of microcrystalline cellulose, with a similar appearance to the active capsules.

Participants and providers were blinded to patient intervention assignment; our biostatistician broke the code only for the final analyses without revealing any specific assignment information to others.

Height was measured without shoes against a wall-fixed tape and weight with light clothing and without

shoes on a platform scale with a 1.0 kg subtraction to correct for the weight of the clothing. The body mass index (BMI) was calculated as weight/height (kg/m^2).

Peripheral blood sample was collected after a 10 hour fasting period from each subject for biochemical parameters, including fasting glucose, lipid profile, fasting insulin, HbA1C, hsCRP and total antioxidant capacity (TAC) at baseline and at the end of the study. Blood glucose was measured using the glucose peroxidase method with the auto analyzer device (Echoplus, Italy). HbA1c was measured by using a chromatography method. Total cholesterol, HDL cholesterol and triglycerides were measured using the enzymatic methods including cholesterol oxidase and glycerol oxidase with the auto analyzer (Echoplus, Italy). Fasting insulin and hsCRP in serum were measured using the ELISA method (Dia Metra, Italy). Total antioxidant capacity (TAC) was determined with a method developed for the evaluation of this parameter in blood plasma. The assay is based on the ability of antioxidants in the sample to inhibit the oxidation of ABTS to ABTS $\bullet$ by a peroxidase. The amount of ABTS $\bullet$ produced can be monitored by reading the absorbance at 734 nm. The assay was conducted at 37 °C to be similar to physiological conditions. Temperature was controlled by a thermoelectric controller probe model CE 2004, Cecil Instrument Ltd, United Kingdom. HOMA Calculator ver. 2.2 (University of Oxford) by analyzing the two parameters fasting glucose and fasting insulin: insulin sensitivity (%S) and HOMA (insulin resistance), which is the reciprocal of %S ($100/\%S$), were measured.

The phenotypic neuropathy assessed in this trial was sensorimotor distal symmetric polyneuropathy, which was assessed by two types of measurements: Physical assessments and nerve conduction study (NCS) using the electromyography machine (Sierra Wave Caldwell Company) at the onset and end of the trial. The indices for physical assessments included deep and superficial sensation assessments, muscle strength and deep tendon reflexes (DTR). All assessments were performed on both sides of the body. Superficial sensation included pain and temperature. Pain (pin prick) was assessed using a sterile needle for determining the length of abnormal area from the toe to the knee. Temperature was assessed by a cool glass and measuring the length of the unfeeling area from the toe to the knee. The deep sensation assessments included joint position and vibration. Joint position was assessed by moving the terminal phalanx of the great toes and coding the patient's feeling of the joint position. Vibration was assessed using a 128 diapason and measuring the length of the unfeel-

ing area from the toe to the knee. Reflex assessment (DTR) of the Achilles tendon was scored as 2 (normal), 1 (decreased), or 0 (absent). Muscle strength was scored as 5 (normal), 4 (good), 3 (fair), 2 (poor: gravity eliminated), 1 (trace: no joint motion produced) and 0 (no muscle contraction). It is notable that we measured the length of the unfeeling area from the toes to the knees in order to assess the progression of the diabetes neuropathy after the trial. The temperatures and conditions used for the assessment were the same before and after the trial. Electrophysiological tests included: Deep peroneal nerve (DPN) velocity, sural nerve action potential (SNAP) amplitude and H-reflex. In the DPN nerve conduction study (NCS), proximal and distal stimulations were performed at the fibular neck and ankle, respectively. The indices were recorded from the extensor digitorum brevis muscle. Sural NCS was performed by stimulation of the nerve trunk at a distance of 14 cm from the lateral ankle border where the recording electrodes were placed. A visual analogue scale (VAS) was used to compare the percent of improvement of neuropathy symptoms after the trial. Each patient was asked to give a number from 0–10 according to the symptoms of neuropathy that they felt (0 = no symptoms to 10 = intolerable symptoms) before and after the trial $[(\text{VAS}_2 - \text{VAS}_1) \times 100]$.

In order to investigate variations in their food intake and to control diet-related confounding factors, three 24 h dietary recalls were recorded from the patients before and after the trial. The average intake was calculated for each macro- and micronutrient before and after the intervention. The Food Processor II software (ESHA Research, Salem, Oregon, USA) was used to process macronutrient and micronutrient intakes based on the dietary reference intakes. The physical activity was assessed by the Persian version of the International Physical Activity Questionnaire (IPAQ) before and after the trial.

With a sample size calculation, we expected that the change in the level of TAC would be $0.5 \mu\text{mol/L}$ after the coenzyme Q10 intervention; hence, the desired power was set at 0.8 to detect a true effect. At an alpha value equal to 0.05 and $S = 0.7$, a minimal sample of 30 in each intervention group and assuming any sample loss, 35 patients were collected in each group. Data were analyzed with the SPSS statistical software. The distribution of the data was evaluated by the Shapiro wilk test. Frequencies of categorical data were analyzed using the Chi-square test or Fisher's exact test, when appropriate. The independent T test (2tailed) was used to analyze the mean changes between groups, while the paired T test was used for within-group

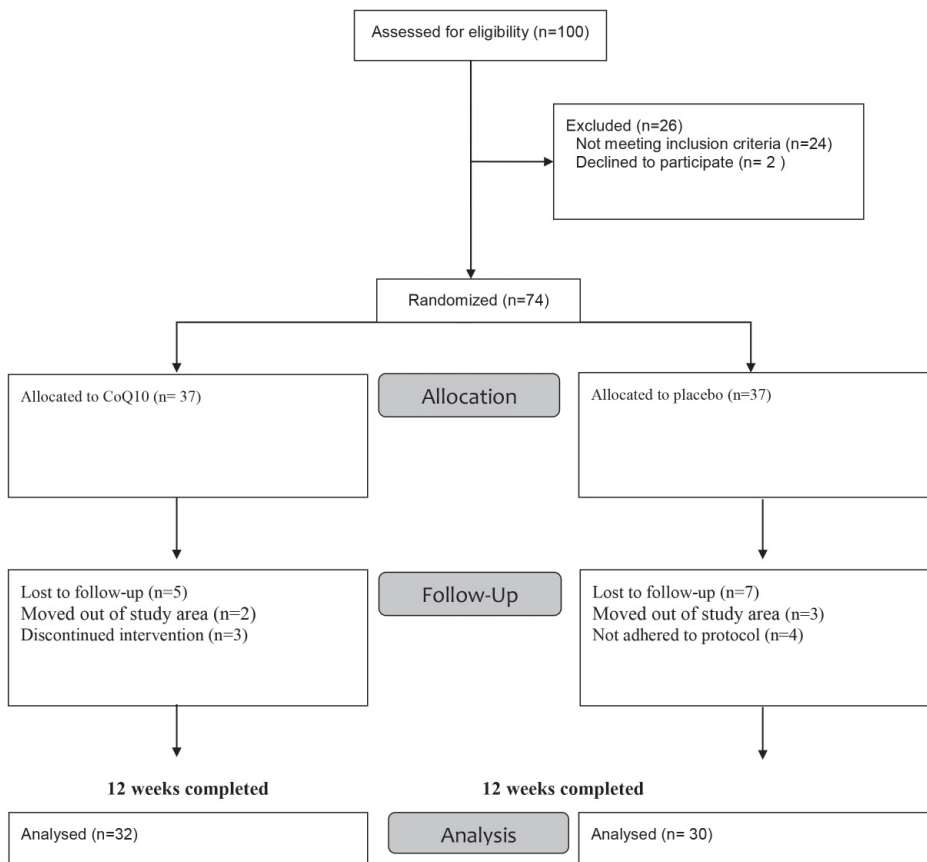


Figure 1: CONSORT flow diagram for studying the effect of CoQ10 on diabetic neuropathy in patients with type two diabetes.

analyses after intervention for normal data. For data which were not normal, the Mann Whitney test was used to analyze the median changes between groups. Log transformation was used for some non-normal distributed data. Adjustment was performed by ANCOVA test considering the baseline concentration as a covariate for normal distributed data.

Table I: Baseline characteristics of participants of the CoQ10 trial.

Variable	CoQ10 (n=32)	P (n=30)
Age (y)	56.7 ± 6.4	54.8 ± 6.7
Male gender (n, %)	10 (31.25)	6 (20)
Weight (kg)	75.7 ± 10.3	77.0 ± 10.6
BMI (kg/m ²)	28.7 ± 4.1	29.6 ± 3.1
Duration of diabetes (y)	16.3 ± 7.3	16.2 ± 7.2
Onset age of diabetes (y)	40.7 ± 8.1	38.4 ± 8.5
Use of oral hypoglycemic agent (%)	9 (28.1)	9 (30)
Insulin users (%)	23 (71.9)	21 (70)

Data are mean ± standard deviation or number (%).

Results

The baseline characteristics of participants are given in Table I. Subjects who received CoQ10 were not statistically different from the placebo group with regard to age, weight, BMI, duration of disease and gender at onset of the trial. Of the 62 participants, 18 were taking oral hypoglycemic agents and 44 were taking insulin. The two groups were similar in all of the observed variables after randomization. Both the CoQ10 capsules and placebo were well-tolerated, and the overall adherence was 96 % during the trial. Pre-post dietary intakes of energy, fat, protein, carbohydrate, and some antioxidant vitamins such as vitamin C, E are featured according to intervention groups (Table II). No significant differences were observed between groups over time. Likewise, no differences were observed for physical activity.

Participants in the CoQ10 group revealed a significant increase in total antioxidant capacity after the trial ($P < 0.001$). There was a significant decrease in hs-CRP in the CoQ10 group which indicated a significant difference between groups after the trial ($P = 0.03$). A significant decrease in insulin sensitivity

Table II: Dietary intake and physical activity levels of participants of the CoQ10 trial.

Variable	CoQ10 (n=32)		Placebo (n=30)	
	Week 0	Week 12	Week 0	Week 12
Energy (kcal/d)	1853.5 ± 115.9	1723 ± 105.0	1835.4 ± 120.8	1805 ± 110.0
Carbohydrate (g/d)	485.8 ± 46.5	466.0 ± 51.0	501.2 ± 48.7	482.0 ± 52.0
Protein (g/d)	58.4 ± 16.7	57.5 ± 12.7	62.3 ± 18.3	61.2 ± 15.3
Fat (g/d)	48.4 ± 16.6	46.2 ± 13.4	46.3 ± 12.6	45.1 ± 11.7
Vitamin C (mg/d)	46.0 ± 5.6	47.0 ± 4.3	48.2 ± 4.7	47.0 ± 3.8
Vitamin E (mg/d)	9.4 ± 1.1	8.9 ± 0.9	8.9 ± 1.3	8.6 ± 1.2
Physical activity (Mets /week)	88.2 ± 30.2	87.5 ± 29.8	85.2 ± 27.2	85.9 ± 25.9

Data are presented as mean ± standard deviation.

Table III: Biochemical parameters before and after 12 weeks of CoQ10 supplementation.

	CoQ10	Placebo	P value*		CoQ10	Placebo	P value*
FBG(mg/dl)				LDL-c (mg/dl)			
Before	166.2 ± 48.3	163.6 ± 51.6	0.8	Before	105.3 ± 21.9	108.6 ± 25.5	0.8
After	157 ± 58	170.3 ± 44.8	0.18	After	105.5 ± 25	109.1 ± 21	0.3
P-value	0.2	0.4		P-value	0.5	0.5	
HbA1c (%)				HDL-c (mg/dl)			
Before	9.05 ± 1.9	9.6 ± 1.6	0.1	Before	32.1 ± 9.9	33.6 ± 7.1	0.5
After	8.7 ± 1.8	9.4 ± 1.6	0.4	After	29.9 ± 4.7	33.0 ± 9.02	0.09
P-value	0.2	0.3		P-value	0.1	0.7	
Insulin sensitivity (%)				TAC (μmol/l)			
Before	88.5 ± 71	78.7 ± 53.6	0.5	Before	7.79 ± 1.99	8.23 ± 2.06	0.3
After	100 ± 81.4	59.56 ± 45.5	0.01	After	9.04 ± 2.02	8.5 ± 1.41	0.8
P-value	0.3	0.04		P-value	<0.001	0.2	
**HOMA-IR				**CRP (μg/ml)			
Before	2.24 ± 2.16	2.15 ± 1.71	0.8	Before	3.77 ± 4.47	3.49 ± 3.74	0.7
After	2.11 ± 2.05	3.33 ± 3.87	0.01	After	2.65 ± 2.81	3.62 ± 3.47	0.03
P-value	0.38	0.027		P-value	0.02	0.09	
Total Cholesterol (mg/dl)				**Fasting Insulin (μIU/ml)			
Before	174.8 ± 34.9	176.4 ± 38.7	0.8	Before	16.18 ± 17.41	14.64 ± 12.57	0.7
After	179.7 ± 31.1	181.4 ± 32.9	0.8	After	15.71 ± 18.23	17.76 ± 13.64	0.02
P-value	0.2	0.4		P-value	0.18	0.04	

Data are presented as mean ± Standard Deviation*ANCOVA was used considering baseline data as covariate ** log transformed data were used due to un-normal distribution. FBP, fasting blood glucose; CRP, c-reactive protein; TAC, total antioxidant capacity.

($P=0.04$) and a significant increase in insulin resistance (HOMA-IR) ($P=0.02$) and fasting insulin ($P=0.04$) in the placebo group was revealed after the trial, which showed a significant difference between groups after the trial for these three parameters ($P=0.01$, $P=0.01$, $P=0.02$) (Table III). ($P=0.01$). The mean changes of insulin sensitivity, HOMA-IR and TAC were significant between groups after the trial (Table IV). No significant changes were reported for the lipid profile. The data for neuropathic parameters are classified in Table V, which demonstrates no significant difference between two groups. The results of the VAS showed

that there was no significant difference in the percentage of improvement of neuropathic symptoms in the Q10 group compared to placebo (Q10: 34.4 ± 28.2 vs. placebo: 43.9 ± 30.8 $P=0.2$).

Discussion

CoQ10 is an intermediate molecule of the mitochondrial electron transport chain. It regulates cytoplasmic redox potential and can inhibit oxidative stress [14]. A defi-

Table IV: Mean and CI of changes in biochemical parameters 12 weeks after supplementation with CoQ10 vs. placebo.

Variable	Co Q10 (n=32)	Placebo (n=30)	p-value*
FBG(mg/dl)	-9.10 (-26.71_8.41)	6.64 (-9.91_23.20)	0.1
HbA1c (%)	-0.29 (-0.71_0.20)	-0.21 (-0.61_0.20)	0.8
Insulin sensitivity (%)	12.10 (11.20_36.41)	-19.10 (-37.80_0.41)	0.04
HOMA-IR	-0.13 (-0.55_0.28)	1.18 (-0.27_2.63)	0.02
Total Cholesterol (mg/dl)	4.81 (-4.40_14.12)	5.01 (-7.21_17.30)	0.9
LDL-c (mg/dl)	0.18 (-0.76_8.04)	0.43 (-8.12_9.04)	0.9
HDL-c (mg/dl)	-2.10 (-5.51_1.16)	0.43 (-8.14_9.04)	0.5
TAC (μmol/l)	1.24 (0.56_1.94)	0.32 (-0.31_0.95)	0.04
hsCRP (μg/ml)	-1.12 (-2.15_-0.09)	0.13 (-0.79_1.05)	0.07
Fasting Insulin (μIU/ml)	-0.47 (-4.13_3.18)	3.11 (-0.67_6.90)	0.1

*Student t-test

Table V: Changes in neuropathic parameters 12 weeks after supplementation with CoQ10 vs. placebo.

variable	CoQ10 (n=32)		Placebo (n=30)		Treatment difference (p= value)
	Baseline	12 weeks	Baseline	12 weeks	
Pain (Cm)	19.32±17.12	18.23±24.75	23.25±13.25	24.23±32.32	0.22*
Vibration (Cm)	0.0±14.00	0.0±17.88	0.0±21.50	0.0±22.0	0.3**
Temperature (Cm)	7.75±22.50	6.5±21.75	20.0±29.25	8.0±30.0	0.2**
Strength (score)	5.0±1.0	5.0±1.0	5.0±1.0	5.0±1.0	0.9**
DTR (score)	1±0.0	1±0.0	1±0.0	1±0.0	0.4**
Deep peroneal nerve (DPN) (m/s)	38.98±5.33	39.50±5.27	37.39±6.13	38.41±6.14	0.7*
Sural SNAP (μv)	4.75±8.0	4.25±8.88	5.0±9.50	2.0±10.75	0.4**
H-Reflex (ms)	60.50±65.5	21.50±32.0	33.0±62.0	15.50±31.0	0.6**

*ANCOVA using baseline values as covariate, data are presented as mean ± SD. DTR, deep tendon reflexes; SNAP, sural nerve action; H-Reflex, Hoffmann's reflex.

**Mann Whitney test was used for analyzing the median between groups after trial; Data are presented as median + inter-quartile range.

ciency of CoQ10 can occur in diabetes due to impaired mitochondrial substrate metabolism and increased oxidative stress [7, 15, 16]. Low serum CoQ10 concentrations have been negatively correlated with poor glycemic control and diabetic complications [12, 17].

In diabetes, the beta cells of the pancreas are disposed to extreme oxidative stress which is due to the impaired antioxidant system. CoQ10 is naturally present in all cells. In increased oxidative stress, the amount of antioxidants including CoQ10 is reduced, which causes beta cell dysfunction and leads to impaired glucose and lipid metabolism [18].

Our study did not show any direct improvement in FBS or glycated hemoglobin, but in the control group, the insulin sensitivity decreased and the fasting insulin and insulin resistance increased, which shows a protective effect in our intervention group during the trial.

Several trials have been performed in these fields, with different findings. In a placebo-controlled trial, Hodgson et al. showed that CoQ10 supplementation lowers glycated hemoglobin significantly in the intervention group [19]. Shargorodsky et al. studied a multi-antioxidant capsule containing vitamin C (500 mg), vitamin E (200 IU), CoQ10 (60 mg) and selenium (100 mcg) in patients with multiple cardiovascular risk factors. The results showed a significant decrease in HbA1c and TG but had no influence on FBG and HOMA-IR [20]. In an open-labeled pilot study, Mezawa et al. concluded that supplementation of ubiquinol in subjects with type 2 diabetes, in addition to conventional anti-hyperglycemic medications, improves glycemic control by improving insulin secretion [12]. In the current study, no significant difference in the lipid profile of patients was observed after the trial between two groups. Modi

et al. showed an improvement in lipid and glucose metabolism in diabetic mice. The potential mechanism was a reduction in the peroxidation of lipids [21]. The lipid peroxidation was not assessed in this trial.

Oxidative stress has been considered by many as an explanation for the tissue damage that accompanies chronic hyperglycemia. It has been reported that erythrocytes from diabetic patients contain low levels of the reduced form of GSH, high levels of the oxidized form (GSSG), and a 51 % reduction in the GSH/GSSG ratio [22]. This has led to many reports of experiments designed to assess whether antioxidant drugs and supplements can be used to protect against oxidative stress in models of type 1 and type 2 diabetes. There are limited studies which have investigated the effect of CoQ10 on the antioxidant state and inflammatory biomarkers in diabetes. The current study showed a significant increase in total antioxidant capacity in the intervention group after the trial (within group comparison) and there was a significant decrease in hs-CRP in the intervention group after the trial compared to placebo (between group comparisons). Lee et al. investigated two different dosages of CoQ10 (60 vs. 150) compared with placebo in CAD. After 12 weeks of intervention, the results showed that the inflammatory marker IL-6 decreased significantly in the Q10–150 group. Subjects in the Q10–150 group had significantly lower malondialdehyde levels and those in the Q10–60 and Q10–150 groups had greater superoxide dismutase activities [23].

The findings of our study showed that supplementation with CoQ10 did not improve the signs and symptoms of neuropathy. In contrast to our study, Hernandez-Ojeda et al., using a randomized clinical trial, observed a significant improvement in neuropathic symptoms/impairment scores, sural sensory nerve amplitude, and peroneal motor nerve conduction velocity with 12 weeks of 400 mg/day CoQ10 compared with baseline values [24]. One of the possible reasons for the results may be supplementing different dosages of CoQ10 (200 mg vs. 400 mg). On the other hand, the discrepancy between the results may be due to the longer duration of diabetes and using insulin in most of our participants.

Currently, there are no treatments for neuropathy, other than treating the diabetic condition per se, but elevated oxidative stress is a well-accepted explanation in the development and progress of complications in diabetes mellitus. Increased free radical-mediated toxicity has been documented in clinical diabetes [25] and animal models of this disease [26]. Oxidative stress is one of the most important determinants of the development of peripheral nerve damage in diabetic neuropathy [7]. The elevated level of toxic

oxidants in diabetic state may be due to processes such as glucose oxidation and lipid peroxidation [27, 28]. As a result, there are several clinical trials regarding the effect of dietary antioxidants such as α -lipoic acid and vitamin E on diabetic neuropathy. The results of a meta-analysis showed that treatment with α -lipoic acid (600 mg/day i. v.) over 3 weeks significantly improves both positive neuropathic symptoms and neuropathic deficits to a clinically meaningful degree in diabetic patients with symptomatic polyneuropathy [29]. In the NATHAN 1 trial, the researchers evaluated the efficacy and safety of α -lipoic acid (ALA) over 4 years in mild-to-moderate diabetic distal symmetric sensorimotor polyneuropathy. This trial resulted in a clinically meaningful improvement and prevention of progression of neuropathic impairments [30]. A randomized, double-blind, placebo-controlled trial involving 21 patients with type 2 diabetes and mild-to-moderate neuropathy was performed to investigate the effect of vitamin E on nerve function parameters. Patients received 900 IU of vitamin E or placebo for 6 months. Both median and tibial motor nerve conduction velocity were significantly improved in the vitamin E group compared with placebo; regardless, no significant changes were revealed in the glycemic parameters [31].

Coenzyme Q10 (CoQ10) is another antioxidant and has bioenergetics and anti-inflammatory effects. It has protective effects against apoptosis of neurons [32] and may be considered an adjuvant therapy with which to treat DPN. Beneficial effects of CoQ10 on DPN have been shown in an animal model [33], and prevented neuropathic pain related behaviors. The analgesic effect of CoQ10 may result from anti-oxidative stress and a further decrease of stress-sensitive and pain-related signaling pathways such as MAPK, NF- κ B and TLR4 [34, 35]. However, in some clinical trials with short-term treatment, antioxidants lacked therapeutic effects in diabetes and its neuropathy [3]. This is partly due to the more chronic, severe, and extensive nature of damage to the nervous system in human diabetes [36]. It seems that combination therapy could provide more effective results. Blocking multiple pathway components by using several antioxidants would in turn block multiple causes of oxidative stress and prevent nervous system injury. It is recommended to study the effects of a cocktail of antioxidants in DPN.

The limitations of this study were the small sample size, long duration of diabetes in the subjects, and the short period of intervention, which in particular seems to have less power to change neuropathy measures in this limited time. The strengths of this study were

the use of human participants and accurate follow-up with the control of some confounding factors such as nutrient intake and physical activity.

In summary, the intake of 200 mg/d of CoQ10, may not improve diabetic neuropathy but can reduce insulin resistance, oxidative stress, and inflammation and also increase insulin sensitivity. Thus, future studies should emphasize longer periods of supplementation and larger doses in milder situations of neuropathy, which may increase the bioactive effects of CoQ10.

Acknowledgements

This study was supported by a collaboration of the faculty of Health and Yazd Diabetes Research Center of Shahid Sadoughi University of Medical Sciences as an MSc dissertation. We extend our sincerest thanks to all subjects who participated in the study.

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